



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 5 • STANDARD 6.GR.1

Area of Regular Polygons

Lesson 5-4 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can find the area of a regular polygon by decomposing it into triangles.

Language Objective: I can explain my steps using the words regular polygon, decompose, triangle, and composite.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Regular Polygon

Decompose

Triangle

Composite

Formula



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

A polygon with all sides and all angles equal is a .

2

To break a shape into smaller shapes is to it.

3

A three-sided polygon is a .

4

Made of two or more shapes combined, a figure is .

5

A math rule written with symbols is a .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

You matched each regular polygon to its number of center triangles. What pattern connects the number of SIDES to the number of TRIANGLES, and why?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐

Regular Polygon — A shape where all sides and all angles are equal.

Polígono regular — Una figura donde todos los lados y ángulos son iguales.

☐

Decompose — To break a shape into smaller, simpler shapes.

Descomponer — Separar una figura en figuras más pequeñas y simples.

☐

Triangle — A shape with three sides.

Triángulo — Una figura de tres lados.

☐

Composite — Made by putting two or more simple shapes together.

Compuesto — Formada al juntar dos o más figuras simples.

☐

Formula — A math rule written with symbols.

Fórmula — Una regla matemática escrita con símbolos.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

I split each polygon into ____ triangles.

Dividí cada polígono en ____ triángulos.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: We decompose the hexagon into triangles because we already know how to find a triangle's area.

Evidence: Students explain that splitting the hexagon into 6 equal triangles lets them use the triangle area formula they already know, then combine the parts.

Reasoning: This evidence connects the definition of regular polygon to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **I split each polygon into ____ triangles.**

Dividí cada polígono en ____ triángulos.

- **The number of triangles equals the number of ____ because ____.**

El número de triángulos es igual al número de ____ porque ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **I know because ____.**

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Regular Polygon
2. **Notes 2:** Decompose
3. **Notes 3:** Triangle
4. **Notes 4:** Composite
5. **Notes 5:** Formula
6. **Watch for this mistake:** *A common mistake in Area of Regular Polygons is stopping after finding the area of ONE triangle and reporting that as the total area — or adding the number of triangles instead of multiplying. Remember: total area = (one triangle's area) $\times$ (number of sides), not (one triangle's area) + (number of sides).*
7. **Try It 1:** A. 60 in^2 — *One triangle = $\frac{1}{2} \times 4 \times 6 = 12 \text{ in}^2$. A pentagon has 5 triangles, so total = $5 \times 12 = 60 \text{ in}^2$.*
8. **Try It 2:** A. 120 sq ft — *An octagon has 8 sides, so it splits into 8 triangles. One triangle = $\frac{1}{2} \times 6 \times 5 = 15 \text{ sq ft}$. Total = $8 \times 15 = 120 \text{ sq ft}$.*